

Car price prediction

Research AI Task 6 | Gradient descent from scratch

The notebook now trains one linear regression model, with a log-price target and ridge penalty 10. These settings were selected by the previous training-only search. Removing the general search framework retains the model and its performance.

What the project does

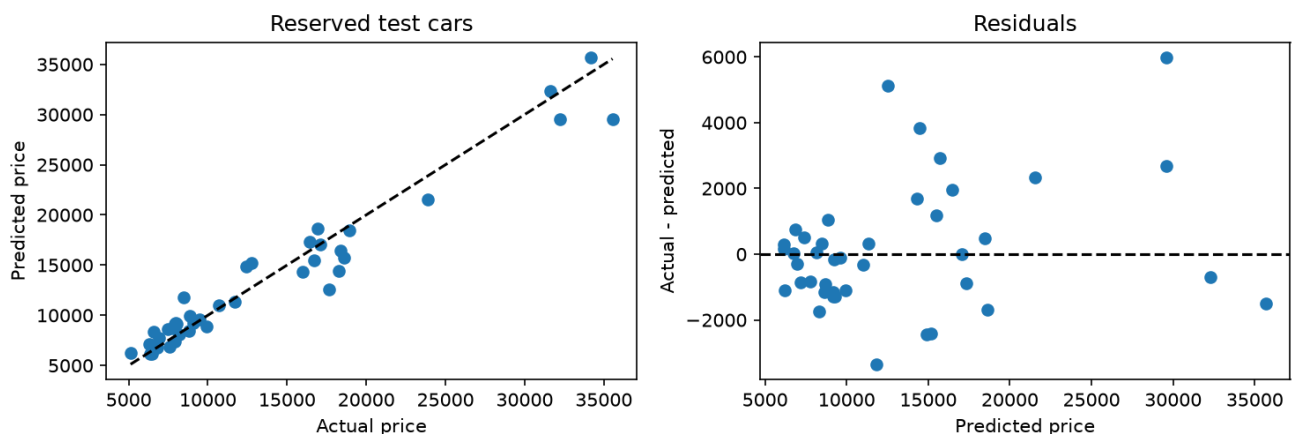
Manufacturer names are corrected, and the row ID and full car name are excluded. The 205 cars are split into 164 training rows and 41 test rows, with identical predictor groups kept together. Encoding, imputation and scaling use training rows only. Five grouped validation folds are repeated three times.

All coefficients are learned through NumPy gradient updates. Momentum speeds convergence; the intercept is not regularized. The model converged in 225 updates. A training-residual correction converts log predictions back to price units. No scikit-learn or direct regression solver is used.

Evaluation metrics

Metric	Training	Cross-validation	Test
MAE	981.99	1,574.94	1,391.31
MSE	2,133,309.13	5,629,657.73	3,706,014.29
RMSE	1,460.59	2,372.69	1,925.10
R-squared	0.9662	0.9109	0.9429
MAPE (%)	7.77	11.95	10.84
sMAPE (%)	7.66	11.73	10.61
Median absolute error	715.45	1,104.76	1,102.80

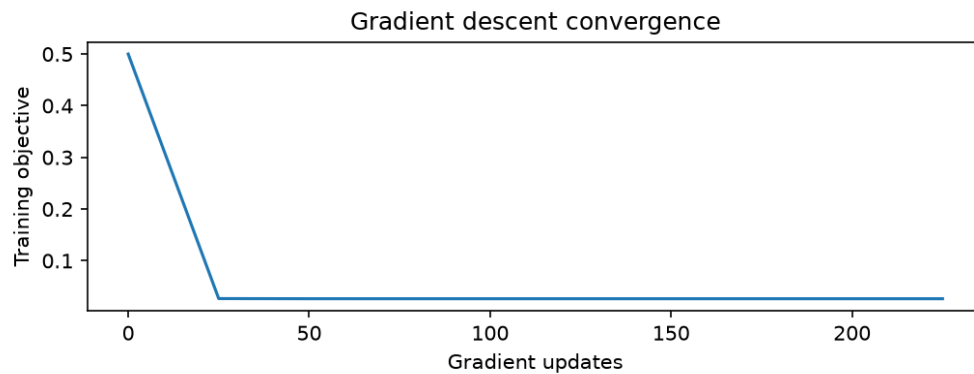
MAE, RMSE and median error use price units; MSE uses squared units. Percentage errors use percent. The supplied dictionary does not identify a currency. Lower errors and higher R-squared are better.



Errors and verification

Test model	RMSE	MAE
Gradient descent	1,925.10	1,391.31
Mean baseline	8,057.91	6,391.57
Median baseline	8,566.58	5,924.37

The selected model reduces test RMSE by 76.1% relative to predicting the training mean. Positive residuals mean underprediction. All cars, including expensive cars, remain in the evaluation.



Largest test errors

Car	Actual	Predicted	Absolute error
jaguar xf	35,550	29,565	5,985
toyota cressida	17,669	12,533	5,136
mazda glc	18,280	14,454	3,826

Verification

The simplified model reproduces the previous predictions within numerical roundoff, with the same reported metrics. Tests cover a known synthetic line, ridge shrinkage, manually calculated metrics, missing and unseen categories, and disjoint data splits. The notebook runs from a fresh kernel.

This small historical dataset cannot establish accuracy for current market prices or unseen manufacturers. Cross-validation variability and the train/test error gap describe uncertainty in these results.

Project repository